CIBC Hackathon: Problem Statement

Design a system for a bank's Collections team that unifies customer data, answers questions in plain English, and powers one AI use case.

Tonight you submit **only the design**. Deadline: **11:00 PM IST, 2 Oct**.

Timeline

When	What
2 Oct, 5:00 to 11:00 PM	System design phase
2 Oct, overnight	Top 50 designs selected
3 Oct, 9:00 AM	Build phase starts, dataset released
4 Oct, 9:00 PM	Build deadline, 10 finalists selected

What you design

Layer	What it does
1. Data Product Factory	Raw sources into one golden customer record (C360), one ID per customer
2. Insight and NLP	Plain-English questions into answers, with the SQL or source shown
3. AI Decisioning	One use case: next best action, agent assist, agent routing, channel optimisation, post-call summary, or QA automation

Governance runs through all layers: access control, monitoring, human review.

What to submit

One PDF per team, max 6 pages, named `TeamName_SystemDesign.pdf`. Team and member names on page 1. No code.

1. **Architecture diagram** (1 page): all three layers, tools named, where AI is used, where a human reviews.
2. **Design doc** (max 4 pages), with these headings:
   - **Summary:** problem, use case, who uses it (3 lines)
   - **Business value:** the metric you improve and how you measure it
   - **Layer 1:** how you build the C360 and match customer IDs across sources
   - **Layer 2:** how questions become answers, and how you refuse out-of-scope ones
   - **Layer 3:** features, model or agent, how each decision is explained
   - **AI usage:** where you use AI and why
   - **Scalability:** what runs in batch vs real time
   - **Governance:** access control, human review, no protected attributes, hardship flags
   - **Build plan:** what you will build in 36 hours if selected
3. **Draft data contract** (1 page): owner, schema, 3+ quality rules, allowed uses, refresh frequency.

Data

You don't need the dataset tonight. Design from the attached data dictionary (31 tables, fully synthetic). Where you can't confirm a field, write "TBC".

Judging

Criterion	Weight
Business impact	25%
Technical quality	25%
Smart use of AI	20%
Clarity	20%
Governance and responsible AI	10%

Rules

- Teams of 1 to 4.

- Explain every decision to the agent.
- A human can review anything that affects a customer.
- No protected attributes in decisions. Flag financial hardship.